Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

Sudhir Dixit[a]

[a] *Independent Researcher*

Abstract

What a recorded field is worth to a prediction is reported as one number, but that number is a functional of rarely stated choices: baseline, pipeline, target, split, decision time, register quality, metric and operating point. We define incremental predictive performance as a *specification surface* $V_s(f)$ over a declared design space under a declared measure, and give four objects for reporting one: an exact functional-ANOVA decomposition of its variance; simultaneous max-t bands over the *whole* surface a claim ranges over, from a pipeline-refitting moving-block bootstrap; *resolution regions* with a robustness index; and *specification regret*. On 19 log–target pairs from 13 public event logs, over an audited 29,040-cell space, the baseline alone moves the increment by 0.073 AUC at the median pair, and a one-number report misstates the sign of 7.5% of resolved cells — 3.5% on the 8 maintained registers. Of the 13 pairs resolving any sign, 3 resolve both. On 40 of 118 models a threshold is not a cost ratio at all. In a case study on 45,455 bank incidents, a register's value turns on the decision time and its sign on the target.

Keywords: specification surface, incremental predictive performance, predictive process monitoring, simultaneous inference, sensitivity analysis, configuration management database

Highlights

- A field's incremental predictive performance is a surface, not one number

Email address: sudhir.dixit1@gmail.com (Sudhir Dixit)

- An exact variance decomposition says which analyst choice moves the answer
- Coverage-calibrated bands say where the sign of that surface is resolved
- Sign misstated for 7.5% of resolved cells, 3.5% on ITSM
- A register's worth turns on the decision time and on which target is set

1. Introduction

A claim that a recorded field is worth something is a comparison. Its second arm — performance *without* the field — is a choice the analyst makes, and so are the learner that makes the comparison, the encoding it uses, the outcome it predicts, the way the data are split in time, whether the field is available at the moment the decision is taken, how completely the register behind the field is populated, the metric the difference is read in, and the operating point at which that metric is evaluated. Each is routinely left implicit, and the result is reported as one number.

This paper's position is not that such a number depends on those choices — that is known, and Section 2 says by whom — but that the dependence is large enough, on real decisions, to change the sign of the answer; that the right object to report is therefore a surface over a *declared* design space rather than a point; and that a surface admits estimation and inference of its own.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) \;=\; m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}),\, D_{\text{test}}) \;-\; m_\theta(A_s(D_{\text{train}}, B),\, D_{\text{test}})\,, \quad (1)$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q, m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one.

We say *incremental predictive performance* and not *value*: (1) measures how much better a model predicts, not what the field costs to acquire or maintain, whether acting on the prediction changes the outcome, or what an organisation would pay for either. Section 8 converts the increment into net

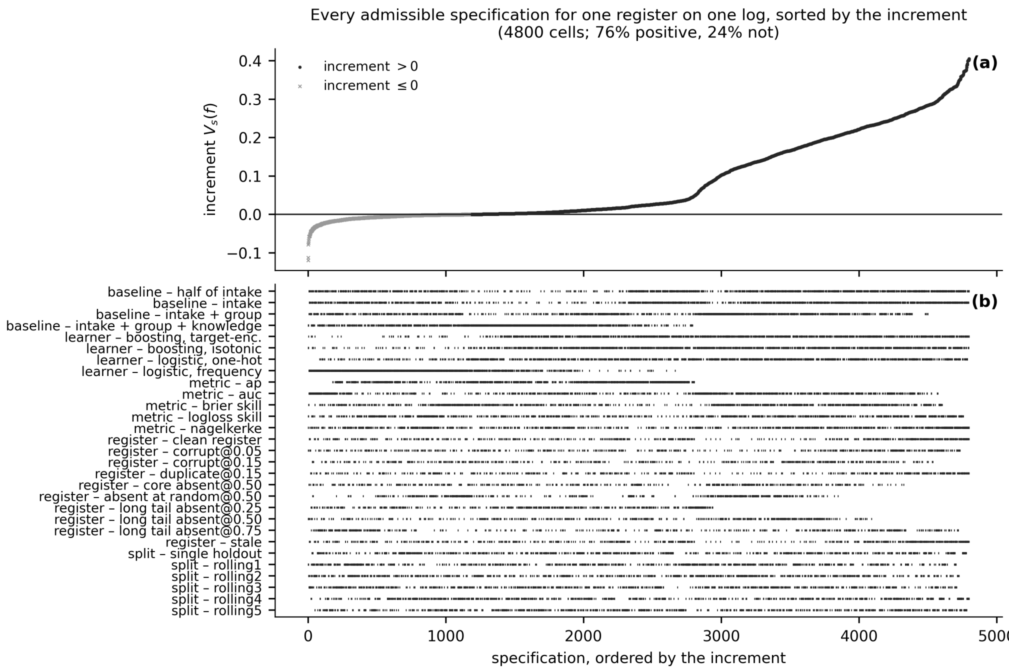

Figure 1: The specification curve for the configuration item on BPIC14, at the registered handover target. **(a)** Each point is one admissible specification — a complete assignment to the pipeline, the baseline rung, the register-quality condition, the split and the metric — sorted by the increment. **(b)** Which level of which axis each specification occupies. The intercept-only rung is excluded. Every point is a point estimate; the simultaneous bands that decide the region labels are in Table S6. **The five scalar instruments are plotted on one axis and their increments are not on a common scale** (Section 4.7); what the panel supports is the *sign* statement, which is unit-free, and not a comparison of magnitudes across instruments. The panel shows the increment rather than the reduction, so a cell where the register is worth almost nothing looks the same as one where the free field absorbs almost everything.

benefit at a declared exchange rate, which is as close to economic value as this evidence gets.

Contributions.

1. **An estimand and four reporting objects** (Table 1): the surface over a declared design space, an exact functional-ANOVA decomposition of its variance, simultaneous max-t bands from a pipeline-refitting bootstrap, *resolution regions* with a robustness index ρ, and *specification regret.*

2. **What each is worth, measured rather than asserted.** Section 10.4 measures the band's family-wise coverage against a known answer and finds it short of nominal; Section 6.5 reports where a specification curve's verdict and a region label part company.
3. **A registered multi-log benchmark.** The surface is computed through one frozen pipeline on every log–target pair a pre-registered protocol admits — 19 of them, not only the ones where the answer resolves — over an audited denominator, and every corpus statistic is reported on all 19 and on the 8 where the register is a maintained one (Section S10.16).
4. **A register at three decision times.** Section 7 shows that a decision time fixes which cases are in scope, what is known about them and which fields are admissible at once, reconciles the two values this log has been reported to give, and finds the difference is the *target*: over the cohort $\times$ target design the target carries 99.3% of the variance and the cohort 0.0%.

What this paper is not. This is a retrospective study on public benchmarks, offered as an evaluation methodology. It is not a procurement conclusion about configuration management databases, not a systematic review of reporting practice, and not a report of a deployed system; Section 11 says what each of those would have required.

The supplementary document. The protocol and its amendments, the proofs, the constructions of the four reporting objects, the correction register, the results this article summarises and the tables it does not print are in a supplementary document whose sections, tables and figures carry an S. A reference here to Section S3 or Table S12 is a reference to that document; the article is self-contained without it.

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.5	S5	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	S6	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.6	3	a function of the band; no further refit
Specification regret	what choosing one specification costs against the best one, on a scale a decision can use	4.7	S16	exact; no refit, on the surface already computed

Table 1: The four reporting objects this paper supplies, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

2. Related work

Every component used here is established. That a feature's apparent importance depends on what else is in the model motivates conditional and permutation importance [33, 12], Shapley attributions [22, 7] and nonparametric formulations of importance as a population quantity [42]; the clinical-prediction literature has said for two decades that the increment in AUC from adding a predictor depends on the existing model [25, 6], is a poor guide to clinical usefulness [6, 37] and is routinely misread [19]. Decision-analytic evaluation supplies the exchange rate [39, 40, 38]. Multiverse and specification-curve analysis supply the sweep [31, 29, 24], underspecification and variance studies the observation that pipelines differ [8, 2], and data valuation a different question about a datum's worth [13, 16]. On event logs, outcome-oriented predictive monitoring supplies the task [35, 36, 28], encoding studies one axis [21, 3], and data-quality work the vocabulary for a degraded register [34, 23, 5]. Leakage discipline is standard [18, 17, 41]. Supplement S1.1 sets out each strand and what this paper takes from it.

The novelty gap, precisely. Every component used here is established and none is claimed. What none of them supplies is the object this paper reports, and the gap is in four places. *The estimand is not declared completely*: the axes are discussed one at a time, and the decision time and the register's quality as preliminaries rather than as arguments. *The measure is left implicit*: a specification curve is summarised uniformly over whatever cells were enumerated, a choice that moves the answer (Section 4.5). *Inference is pointwise or absent*: a curve is a distribution of point estimates and a permutation test on its median, which licenses no statement about a sub-region. *Nothing measures what the conventional report costs*: the alternative to one number is offered as a display, not as a decision rule with a loss. Of the four objects that close these, the decomposition and the band are adaptations; *resolution regions* with a robustness index and *specification regret* are, as far as we are aware, new. Section 6.5 measures where a curve's verdict and a region label part company rather than asserting a difference.

The contribution is deliberately *not* "people usually report one number". It is that a claim of this form is a functional of a design, that the functional can be estimated and its uncertainty quantified, and that on this corpus it is far from constant.

3. The specification surface

3.1. The design space

Definition 1 (Design space)**.** *A* design space $\mathcal{S}$ *for a feature-value claim is a finite set of specifications, each a complete assignment to the axes*

$$s = \Big(\underbrace{A}_{\textit{learner}}, \underbrace{E}_{\textit{encoding}}, \underbrace{Y}_{\textit{target}}, \underbrace{\Pi}_{\textit{split}}, \underbrace{\tau}_{\textit{decision time}}, \underbrace{q}_{\textit{register quality}}, \underbrace{B}_{\textit{baseline}}, \underbrace{m}_{\textit{metric}}, \underbrace{\theta}_{\textit{operating point}} \Big),$$

together with the cohort and eligibility rule that fix D. A claim about a feature is well posed *only relative to a declared $\mathcal{S}$.*

axis	levels	measure	exclusions
learner	2-4	equal / ref. / conc.	none
encoding	2-3	set by the learner	none
split	6	equal / ref. / conc.	none
decision time (availability)	1-2	never averaged	two times on the primary log, one elsewhere
register quality x severity	6-10	equal / ref. / conc.	none
baseline (ladder rung)	4-5	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3-4	equal / ref. / conc.	none
scalar instrument	5	never averaged across	none
operating point (net benefit)	31	declared threshold dist.	outside the declared range

Table 2: The declared design space. One row per axis: the number of levels (a range where it differs by log), the measure the paper places on those levels, and every level excluded by declaration with its reason. Generated from the code that walks the space, so the level counts multiply to the cell counts of Table S12.

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level that is deliberately excluded with the reason. Table 2 is that declaration for this study, and it is generated from the code that walks the space rather than typed beside it. A reader can multiply its level counts and obtain the cell counts reported in Section 6; that this multiplication works is the first check the verification harness runs.

The measure placed on the axis levels is a modelling choice and is declared rather than left implicit; Supplement S1.2 states it, and says why the decision time and the register's quality are axes of the estimand rather than preliminaries to it.

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test}, so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. That is not the quantity this paper reports intervals for, and the distinction has to be made before Section 4 can say what its bootstrap targets.

Definition 2 (The estimand)**.** *Let the cases of a log be a sample from a population P, and let the specification s fix the fitting procedure A_s, the cohort and eligibility rule, the split rule Π, the decision time, the register-quality mechanism, the baseline and the instrument. Write $A_s(P)$ for the limit of the fitted scoring rule as the sample grows under P with s held fixed. The* estimand at *s is*

$$\vartheta_s(f) \;=\; m_\theta\Big(A_s(P, B \cup \{f_q\}),\, P\Big) \;-\; m_\theta\Big(A_s(P, B),\, P\Big),$$

and $V_s(f)$ of (1) *is its plug-in estimate on the log.*

Three things follow and are used throughout. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away: ϑ_s is what this analyst's procedure converges to, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it, and why the split is an axis. Under misspecification ϑ_s is not what the register is worth in the world — the simulation of Section 10 separates the two and measures the gap, calling them V_{limit} and V_{oracle}. And every cell of every figure and table in this paper is an *estimate* of ϑ_s; "resolved" in Definition 3 means that the band excludes zero for ϑ_s, not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D, the surface is the map $s \mapsto V_s(f)$ of (1), and a conventional report is one cell of it. The paper's claim is that a single cell is not a summary of the surface in any useful sense: Section 6 shows that the sign of V_s disagrees across cells on most log–target pairs we study, and Section 6.4 measures how often. There is one state in which a scalar is defensible: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive number is a lower bound on a robust conclusion. Section 4.6 makes that a definition.

3.4. The relative reduction, and its pathologies

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) \;=\; V_s(f \mid B_0) - V_s(f \mid B_1), \quad R_s(f) \;=\; 1 - \frac{V_s(f \mid B_1)}{V_s(f \mid B_0)} \;=\; \frac{D_s(f)}{V_s(f \mid B_0)}. \tag{2}$$

D is an absolute difference in the metric's own units. R is a ratio and inherits every pathology of one: the denominator can be small, zero or negative; R is not comparable across metrics, because Proposition 1 says the two are not functions of each other; and R can exceed 1 or fall below 0, so reading it as "the share absorbed" is a category error whenever it leaves $[0, 1]$.

We therefore make D and the two increments primary throughout and treat R as a secondary descriptive statistic subject to three rules, declared before any R in this paper was computed:

1. R is reported only where $V_s(f \mid B_0)$ is resolvably positive under the *simultaneous* band, not merely positive as a point estimate;
2. its interval is a Fieller set and its *kind* — bounded, unbounded, or the complement of an interval — is printed beside it. A confidence set for a ratio is the solution set of a quadratic [11] and is a bounded interval only when that quadratic's leading coefficient is positive, which is to say only when the denominator is resolvably away from zero; otherwise it is unbounded or is the *complement* of an interval, and no bounded interval summarises it (Supplement S3). Of the 10,584 reductions this paper computes, 5,282 have a non-bounded Fieller set, and a percentile interval reports a bounded one for every single one of them;
3. the magnitude of R is never compared with the spread of R across metrics as though both lived on an unproblematic common scale.

3.5. Two propositions

Throughout, a *configuration* is a quadruple of score vectors on a fixed labelled test set, $\mathcal{C} = (\sigma_{B_0}, \sigma_{B_0 \cup f}, \sigma_{B_1}, \sigma_{B_1 \cup f})$ with $B_0 \subset B_1$, and $V_m(B)$, $R_m(\mathcal{C})$ are as in (1) and (2). The metric conventions the constructions below depend on — the tie rule for ROC AUC, the step-wise definition of average precision, the selection rule for net benefit — are fixed and listed in `results/s07_conventions.csv`.

Proposition 1 (Metric non-identifiability of the reduction)**.** *There is no function $\varphi : \mathbb{R} \to \mathbb{R}$ with $R_{\mathrm{AP}}(\mathcal{C}) = \varphi\Big(R_{\mathrm{AUC}}(\mathcal{C})\Big)$ for every configuration $\mathcal{C}$ whose denominators are positive under both metrics. Consequently the reduction is not identified by the data, the feature and the two baselines: the metric is an argument of the estimand.*

The construction is in Supplement S2.

It establishes non-identifiability and not unboundedness: R is built over a bounded interval in any fixed configuration, so the reading is that no function carries one metric's reduction to another's. In the two configurations of `results/s29_prop1.csv` the AUC reduction is identical to 2.2×10^{-16} while the average-precision reduction differs by 0.417.

Assumption 1 (Richness)**.** Every quadruple of score vectors on the fixed labelled test set is realisable as a configuration. This is the assumption that no restriction is placed on the model class; it is what makes "for every configuration" quantify over a set with more than three elements, and it fails only if the analyst commits in advance to a family of scoring rules narrow enough to constrain the four arms jointly.

Proposition 2 (What survives a declared family of recalibrations)**.** *Let Ψ be a family of strictly increasing continuous maps containing the identity, each applied* simultaneously *to all four score vectors of a configuration — one recalibration of the scoring system, not a different one per arm. Call m* Ψ-invariant *if $m(\psi(\sigma), y) = m(\sigma, y)$ for every $\psi \in \Psi$ and every (σ, y). Under Assumption 1, the following are equivalent.*

(i) $R_m(\psi\mathcal{C}) = R_m(\mathcal{C})$ *for every configuration $\mathcal{C}$ with nonzero denominator and every $\psi \in \Psi$.*
(ii) *For every $\psi \in \Psi$ there are constants $A(\psi) \neq 0$ and $B(\psi)$ with $m(\psi(\sigma), y) = A(\psi)\, m(\sigma, y) + B(\psi)$ for every σ.*

Richness is used only for (i) $\Rightarrow$ (ii); the direction that matters in practice needs none. Taking Ψ to be every strictly increasing continuous map recovers the unrestricted statement.

Within a declared family, invariance is strictly weaker than rank-basedness. A rank-based instrument satisfies (ii) with $A \equiv 1$, $B \equiv 0$, hence (i); the converse fails for some Ψ. The class-mean gap is a witness: it is not rank-based, an affine recalibration moving it by up to 0.080, yet its reduction is invariant to 1.9×10^{-14} inside the affine family. **The separation is family-relative** — over 5 non-affine families the same witness's reduction shifts by up to 0.638 — so an analyst declares the recalibrations they will consider and the proposition answers for those. **What it buys** is that a search becomes a proof: one ψ and one triple violating (1) is a *certificate* that rules out invariance for every configuration, and `results/s29_certificates.csv` carries 18 of them, the smallest moving R by 0.241, against 2,400 checks in which the two rank-based instruments hold to exactly zero. Supplement S2 gives both propositions' constructions, the role of richness, and the witness in full.

4. Estimation and inference

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log, and the *pipeline* axis spans encodings as well as model families. Attributes are assigned to roles by the pre-registered rules of Section 5, cases are ordered by their first timestamp, and the split is temporal; Supplement S3.2 lists the levels and why the tree learner is target-encoded. The usual interval for such an increment resamples test rows with the models held fixed, which excludes training-sample, model-selection, encoder and split variability and the temporal dependence between cases.

Every interval here is instead a *nested moving-block bootstrap*: one draw resamples the training half by moving blocks [20] of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and quality mechanism, and evaluates on a moving-block resample of the test half, with encoders, frequency tables, target encodings, register rankings and calibrators rebuilt inside the draw. On the simulation of Section 10 the fixed-model interval covers a median 92.1% and the nested basic interval 93.0%.

Refitting inside the draw breaks the percentile interval. A bootstrap resample holds about $1-e^{-1}$ of a high-cardinality categorical's distinct levels, so every refit sees a sparser register and returns a smaller increment; the bootstrap distribution is displaced and a percentile interval taken from it is a correct interval for the wrong quantity. On the sparse world of Section 10 it covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\,2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2}\,]$ [9], and simultaneous bands are recentred by the same shift (Supplement S3.7).

A pivoted interval need not contain the estimate it is built from, and on this corpus 74 cells do not. Pivoting moves the interval by twice the displacement. Where the displacement exceeds the interval's own half-width, both endpoints land on the same side of $\hat{V}$ and the band excludes it. That is a property of a pivotal interval under bias rather than an error, but it is not what a reader expects of a band, so it is counted: 74 of the 3,900 whole-surface cells — 1.9% — on 3 of the 19 pairs, reaching 21.1% of the cells on the worst of them. The ratio of displacement to half-width has a median 0.09 across pairs and 0.45 on that pair.

It gets worse with the sample size, not better. The displacement is a bias — a resample holds about $1 - e^{-1}$ of the distinct levels however many rows it has — so at a fixed ratio it does not shrink with n while the half-width shrinks like $n^{-1/2}$. Section 10.3 locates the crossing: at a ratio of 0.125 the interval straddles its own estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000. The three pairs above are the corpus's nearest it, and Section 6.3 reports what it costs the coverage calibration.

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* is a statement about a denominator, and the denominator has to be the set the statement ranges over. Three sets are distinguished, named, and counted from the files rather than quoted (Table S12). The **declared surface** is every scientifically admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface — 29,040 scalar cells against 29,040 declared, with 0 duplicates and 0 missing — and the audit script asserts that identity rather than the manuscript asserting it. The **inference surface** is every specification carrying bootstrap draws; it is smaller for one reason, that a draw refits the whole pipeline and its cost is the declared surface's multiplied by the draw count.

The inference surface is declared before any draw is taken, by a rule that reads only a log's row count, and it is *axis-complete*: every axis varies over at least two levels, so no axis is silently held fixed. It carries 3,900 scalar family members, a median 12.5% of each pair's computational surface, and the share of cells with a positive increment differs between the two surfaces by a median 0.067. Every quantity that does *not* need a bootstrap — the decomposition, the sign-disagreement rate, the regret comparison — is computed on the full declared surface; only the region labels are computed on the inference surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a ladder read across rungs, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} \;=\; \max_{c\in\mathcal{F}} \frac{\left|V_c^{(b)} - \bar{V}_c\right|}{\hat{\sigma}_c}, \qquad q_{1-\alpha} = \text{the } (1-\alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha}\hat{\sigma}_c$. This is the max-$t$ (Romano–Wolf) construction [26] and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger.

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells a surface-level label ranges over. Four interval objects are declared and every table says which it prints: POINTWISE, WITHIN-INSTRUMENT, WHOLE-SURFACE — max-t over *every* admissible scalar cell of the pair, the only family a surface-level claim may use — and DECISION-CURVE, declared separately because net benefit and a scalar instrument are not the same scale. The whole-surface family has a median 180 members against five, and its median critical value is 3.33 against 2.35 and the pointwise 1.96. Family control is per log–target pair; every cross-pair statement here is descriptive. Supplement S3.3 gives the four objects and the rule for dropping a replicate that leaves a cell without outcome variation.

Estimating the critical value without paying for it in refits. $q_{1-\alpha}$ is the upper quantile of a maximum over hundreds of correlated cells and the empirical quantile of B observed maxima estimates it badly, while raising B multiplies the cost of the whole design space. Standardising the draws and multiplying them by independent standard normals gives a statistic with their empirical covariance, generated 20,000 times for the cost of matrix arithmetic [4]; Supplement S3 gives the construction. The budget therefore fixes B and not the precision of q — order-statistic widths of 2.01 against 0.037 — and the residual Monte Carlo interval is carried through, so a cell counts as *resolved* only at its *upper* end. A max-t band also inherits the coverage of the interval it is built from, which Section 10.3 shows is governed by the ratio of register cardinality to training size; Section 6.3 calibrates q at each pair's own ratio.

Which of them is right, measured against a known answer. A disagreement between two estimators of one quantity says that one of them is wrong and cannot say which, so Section 10.4 measures the family-wise coverage each one attains over synthetic families whose size, draw count, cross-cell correlation and tail behaviour are matched to this corpus, and whose truth is known by construction. Three results govern how the bands in this paper should be read, and none of them is comfortable.

The multiplier band does not attain its nominal level anywhere in this corpus's configuration, and the shortfall is not a tail effect. Under *Gaussian* draws — the case the multiplier is derived for — its family-wise coverage has a median 87.3% over the matched cells and a minimum 74.8% against a nominal 95%; under draws matched to this corpus's tails, 84.6% and 76.1%. The construction is asymptotic and these draw counts are not asymptotic.

The safeguard the resolved flag uses buys almost nothing. Resolving a cell only at the upper end of the critical value's Monte Carlo interval moves coverage from 84.6% to 84.8%, because that interval is narrow by design — it is the precision the multiplier was adopted for. It controls the Monte Carlo error in q and not the error in what q estimates.

The best of the five candidates is the empirical quantile taken at the upper end of its own order-statistic interval, and it is still short. It reaches a median 92.2% and a minimum 86.5% on the matched cells, at 1.52 times the width. A Rademacher wild multiplier, which reuses the observed draws with random signs instead of replacing them by Gaussians and which we expected to repair exactly this, does worse than the Gaussian one: 80.7%. Section 10.4 reports all five.

The binding constraint is the number of draws, not the choice among estimators. At the modal surface family the multiplier band covers 82.0% at 33 draws, 90.9% at 80, 94.4% at 150 and 95.8% at 400; the per-cell interval it is built from moves 90.3% to 94.6% over the same range. This corpus runs at 40 to 150 draws as declared and 33 to 150 effectively, and it is the effective count a band is built from. Every band, region label and ρ in this paper should be read as the anti-conservative statement that makes it, and Section 11 states what follows.

4.4. Planned contrasts, and a p-value the design can produce

Five contrasts on the primary log are *planned for this analysis round*: that the register adds to the intake baseline (PC1); that it still adds once the free field is admitted (PC2); that it adds at the later decision time (PC3); that the absorption D is positive (PC4); and that a half-empty register is worth less than a full one (PC5). Each is stated with its complete estimand — log, cohort, target, learner, encoding, split, decision time, register quality, baseline and instrument — in Table S13. They are not *confirmatory*: they were fixed before this analysis round, not before the data were seen. The p-values are one-sided bootstrap p-values with a Holm step-down correction [14] at $\alpha = 0.05$, computed with the plus-one estimator $\hat{p} = (k+1)/(B+1)$, where k counts draws on the null side; a p-value of zero is a number no finite bootstrap can produce. At 2,000 draws the smallest attainable Holm-adjusted value over five tests is 0.00250, and 5 of the five survive.

A one-sided p-value and a two-sided basic interval are different statements about the same draws, and on PC3 they disagree; **a contrast is called resolved only when the interval says so**, and PC3 is not resolved on any of the four cohort-and-target combinations Section 7.1 reports. Supplement S3.4 works the disagreement through. Every other comparison here is *exploratory* and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

4.5. The sensitivity decomposition

Because the surface is a complete factorial over its axes — the audit of Section 4.2 establishes that it is, cell for cell — the variance of V_s admits an exact and *orthogonal* functional-ANOVA decomposition [30, 27] under a product measure $\mathbb{P} = \bigotimes_i w_i$. Every subset u of axes has a component g_u, a Möbius sum over the conditional means computed exactly rather than estimated (Supplement S3); the components are orthogonal, so $\sum_{u \neq \emptyset} D_u = \operatorname{Var}(V)$ exactly, and with $S_u = D_u / \operatorname{Var}(V)$ the first-order and total indices are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u \ni i} S_u$. The closest precedent is Hutter et al. [15], who decompose an algorithm's performance over its *configuration* space by the same device; what differs here is that the axes are analysis decisions rather than hyperparameters, that the components are exact over a complete factorial rather than estimated from a surrogate over a sampled design, and that the decomposition is one of four reporting objects rather than the conclusion.

The per-axis difference $S_{T_i} - S_i$ is the variance axis i is *involved* in beyond its main effect. These overlap — a two-way component appears in both its axes' total effects — so their largest member is not the share carried by interactions. That share is

$$S_{\text{int}} \;=\; 1 - \sum_i S_i, \tag{3}$$

which this paper computes together with every disjoint component, so the identity $\sum_u S_u = 1$ is asserted numerically — it holds on all 304 decompositions — rather than assumed. The other quantity is retained as the *largest per-axis interaction involvement*; the two differ substantially, at 43.7% against 56.0% at the median pair.

4.6. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions)**.** *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws, a median 12.5% of that pair's admissible scalar cells. Under the simultaneous band of Section 4.3, a cell of $\mathcal{F}$ is* beneficial *if its lower band is above zero,* harmful *if its upper band is below zero, and* unresolved *otherwise. A surface is* uniformly beneficial *if every cell of $\mathcal{F}$ is beneficial;* conditionally beneficial *if some are and none is harmful;* uniformly harmful *and* conditionally harmful *in mirror image;* sign-changing *if there is at least one cell of each kind; and* unresolved *if none is resolved. The* robustness index *is*

$$\rho \;=\; \frac{\#\{\mathit{beneficial}\} - \#\{\mathit{harmful}\}}{|\mathcal{F}|} \;\in\; [-1, 1].$$

The label is relative to $\mathcal{F}$ and not to the whole admissible set, and the direction of that is not neutral. A label computed on a sixth of a pair's cells is systematically cleaner than one computed on all of them, and *uniformly beneficial* is the label most helped by it: a claim about every cell is easier to sustain over fewer cells. The manuscript therefore never says a surface is uniformly beneficial without this restriction being in force, and Section 6.3 reports that the label is not reached on any pair in any case.
The two harmful labels are not decoration: Table S6 carries 1 conditionally harmful and 0 uniformly harmful surfaces out of 19. $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and the empirical question this paper answers is how often it obtains. The region is defined over the *scalar* family; folding the decision-curve cells into it would let a single extreme threshold decide the label of a whole surface, so the decision curve gets its own region (Section 8). The intercept-only rung is excluded from every admissible set, because a reduction measured against a model with no features is inflated by construction; the master surface file *does* carry that rung, which is why the two counts in Table S12 differ.

4.7. Specification regret

The fourth object asks what choosing one specification costs against the best one, on a scale a decision can use. Three designs are declared and all three reported: **A**, metric-specific, so that no average crosses an instrument; **B**, on calibrated net benefit, which is a common operational utility; and **C**, partially identified over a declared set of monotone utility maps, which reports an interval rather than assuming one map. The identity map's maximum excess is exactly zero. Supplement S3.1 gives the three constructions and the four decision rules compared under them.

5. The registered benchmark design

5.1. What was fixed, and when

The rules that turn an arbitrary event log into an instance of this problem were committed before any result on a new log existed, in `PROTOCOL.md` for the corpus and `AUDIT-PROTOCOL-2.md` for the practice pilot. The version history shows the registering commit adding one file and no result file; `git log --stat` is the check, and `REPRODUCE.md` names the commit. Six amendments were declared afterwards, each with its date, its cause and the result it changed, and all six are in Supplement S1 rather than folded silently into the rules.

This is not a timestamped third-party registration, and the difference matters. A registration on a public registry is attested by a party with no stake in the result; what is offered here is a file in a repository whose history the authors control, so a reader who does not trust that history has nothing else to check. Any claim in this paper that a rule was fixed in advance is therefore a claim about the repository and no stronger than the repository. We say so plainly instead of letting the word *registered* carry a weight it has not earned: what the design buys is that the rules are legible, dated and amendable only on the record, not that an outside party watched them being written.

5.2. Roles

A log is admitted if the registered rules can find three things in it without reading an outcome.

The register f. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field g. A resource-like attribute, present on at least 50% of cases, whose value varies within a case, so that it can carry information about routing.

The intake block B_0. Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f.

Coarsenings of f — its type, its subtype, the service it belongs to — are given their own role rather than being swept into B_0, because which *layer* of a register pays is a separate question, answered in Supplement S4.

5.3. Targets, splits and exclusions

Two targets are defined for every log — *handover*, whether more than one resource group touches the case, and *duration*, whether it runs longer than the training half's median. Cases are ordered by first timestamp and split 70/30, with five expanding-window rolling-origin folds beside it, and fields determined only at closure are excluded by name [18, 17, 41]. A pair is excluded only for a registered reason, and **every admitted pair is analysed and reported** — running the surface only where the reduction resolves would condition the analysis on the outcome. Supplement S1.3 gives the rules, the exclusion codes and the admission margin at the upper edge of the prevalence window.

5.4. What the corpus does and does not have in common

The corpus is 13 public logs spanning 6 domains, and it includes the further ITSM logs the literature uses [32, 1]. It is heterogeneous and the paper does not claim otherwise. The registers are a configuration item, a caller, a product, a vendor, a diagnosis and a legal article; the free fields are an assignment group, an assignee, a resource and a role; the targets mean different things in a bank's incident queue, a hospital's admissions and a municipality's permits.

Table S11 makes that explicit rather than leaving it to a sentence. For every pair it names the domain, the field playing the register role, why that field behaves as a maintained and reused entity, the free field, the interpretation of the target, and the specific threat to construct validity that the pair carries. A diagnosis code is admitted as a register by a cardinality-and-reuse rule, and it is not a configuration management database; a legal article is reused across cases but nobody maintains a discovery process for it. Those are threats, they are named per pair, and the paper's conclusions are stated at a level that survives them.

This is therefore a *benchmark of specification sensitivity across heterogeneous prediction problems*, not a replication of one finding on several organisations, and two disciplines follow. No domain-specific effect magnitude is averaged across domains as though it estimated one universal register value: the corpus-level statistics are all about *sensitivity*, which is a property each pair has separately and which can be summarised across pairs without assuming they measure one thing. And the 8 ITSM-like pairs, the closest family to the case study, are analysed as a group as well, with *every* headline corpus statistic reported on both populations in Section S10.16.

6. Multi-log results

Every point estimate in this section comes from one master surface, `results/s01_surface.csv`, computed by one pass of one pipeline over 19 log–target pairs, 4 learners, 6 splits, 10 register-quality conditions, the nested baseline ladder and 36 instruments including the 31-point decision-curve grid. Its 273,024 rows *include* the intercept-only rung, so that the surface is complete; no admissible set contains it, and the 29,040 admissible scalar cells every headline is computed on exclude it. Every interval is the basic construction of Section 4.1 and every band the whole-surface simultaneous construction of Section 4.3, and each table names which it prints. The denominator audit is in Table S12: declared equals computed on all 19 pairs, with 0 duplicates and 0 missing, and the inference surface (3,900 scalar members) is a subset of it cell for cell, which `verify_numbers` asserts.

6.1. The master table

Table 3 is the frozen master table: one row per log–target pair, the increment at the reference specification with its simultaneous band, the region label, the robustness index and the sign-disagreement rate of a conventional one-number report. The reference specification is declared and is the same everywhere — one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field as the baseline, ROC AUC — and its region and ρ columns are the coverage-calibrated ones of Section 6.3 while its sign-disagreement column is the all-cells rate of Table S7 under the equal-level measure.

6.2. What the choices are worth

Table S5 and Figure 2 give the decomposition, computed within each instrument in that instrument's own units, under the equal-level measure. The table's first four columns are, per pair, the median over the five instruments of each axis's first-order index; its last column is the largest per-axis interaction *involvement*. Three aggregations of this table are quoted below and they are not interchangeable, so each is named where it appears: the median over

log	target	V at reference	pointwise	simultaneous	region	rho	misreport rate
BPIC13_closed	handover	-0.009	[-0.041, +0.018]	[-0.068, +0.046]	unresolved	0.000	0.284
BPIC13_incidents	duration	0.028	[+0.013, +0.060]	[-0.003, +0.074]	conditionally beneficial	0.017	0.393
BPIC13_incidents	handover	0.037	[+0.030, +0.050]	[+0.021, +0.059]	conditionally beneficial	0.422	0.111
BPIC14	duration	0.096	[+0.089, +0.109]	[+0.081, +0.119]	conditionally beneficial	0.692	0.106
BPIC14	handover	0.167	[+0.143, +0.205]	[+0.115, +0.228]	sign-changing	0.552	0.244
BPIC15_1	duration	0.001	[-0.029, +0.046]	[-0.073, +0.071]	conditionally beneficial	0.011	0.423
BPIC15_1	handover	0.007	[-0.007, +0.021]	[-0.017, +0.032]	unresolved	0.000	0.242
BPIC15_3	duration	0.035	[-0.000, +0.070]	[-0.025, +0.094]	conditionally beneficial	0.056	0.191
BPIC15_3	handover	-0.003	[-0.047, +0.058]	[-0.088, +0.088]	unresolved	0.000	0.616
BPIC15_4	duration	0.035	[+0.016, +0.084]	[-0.008, +0.101]	unresolved	0.000	0.623
BPIC15_4	handover	0.001	[-0.026, +0.056]	[-0.059, +0.076]	unresolved	0.000	0.618
BPIC15_5	duration	0.032	[-0.026, +0.100]	[-0.075, +0.143]	unresolved	0.000	0.328
BPIC17	duration	0.002	[-0.003, +0.010]	[-0.008, +0.018]	conditionally beneficial	0.139	0.353
BPIC19	duration	0.024	[+0.011, +0.043]	[-0.001, +0.052]	sign-changing	-0.083	0.360
Helpdesk	duration	-0.020	[-0.039, -0.003]	[-0.054, +0.007]	conditionally harmful	-0.011	0.257
RoadFines	duration	0.031	[+0.025, +0.037]	[+0.020, +0.042]	sign-changing	0.225	0.173
Sepsis	duration	0.153	[+0.114, +0.210]	[+0.076, +0.237]	conditionally beneficial	0.283	0.056
UCI498	duration	-0.007	[-0.011, +0.006]	[-0.016, +0.015]	conditionally beneficial	0.094	0.493
UCI498	handover	-0.003	[-0.007, -0.001]	[-0.009, +0.002]	conditionally beneficial	0.044	0.481

Table 3: The master table. One row per admitted log–target pair. 'pointwise' is the 95% interval at the reference cell, basic (pivotal) construction; 'simultaneous' is that same cell's WHOLE-SURFACE simultaneous band, at the conservative end of the critical value's Monte Carlo interval. They are different objects, and the region label uses only the second. Then the resolution region and the robustness index ρ, both under the COVERAGE-CALIBRATED critical value of Section 6.3 — the nominal ones are in Table S6 — and the share of admissible specifications whose sign disagrees with a conventional one-number report, under the equal-level measure, which is the 'all-cells rate' column of Table S7. The reference cell's sign and the region need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative.

pairs of a column of this table; the 28.7% figure, which is the median over pairs of the median over instruments of the *per-instrument* largest first-order index, and which is therefore not obtainable by taking a maximum across the four printed columns; and the corpus medians of the higher-order share. The per-instrument quantities the second is built from are in Table S24.

No single axis dominates, and the largest is the one an evaluation paper is most likely to fix without comment. Across the corpus the median first-order index is 18.9% for the split, 5.5% for the baseline rung, 4.2% for the register-quality condition and 3.6% for the pipeline. The split leads the other three by a wide margin and still explains well under a quarter of the variance. At the median pair the largest of the four, whichever it is, is 28.7%.

Most of the variance belongs to no single axis. The total higher-order share $1 - \sum_i S_i$ has median 56.0% (95% interval 40.9% to 60.8% over pairs), ranging from 23.3% to 79.3%; the largest per-axis *involvement*

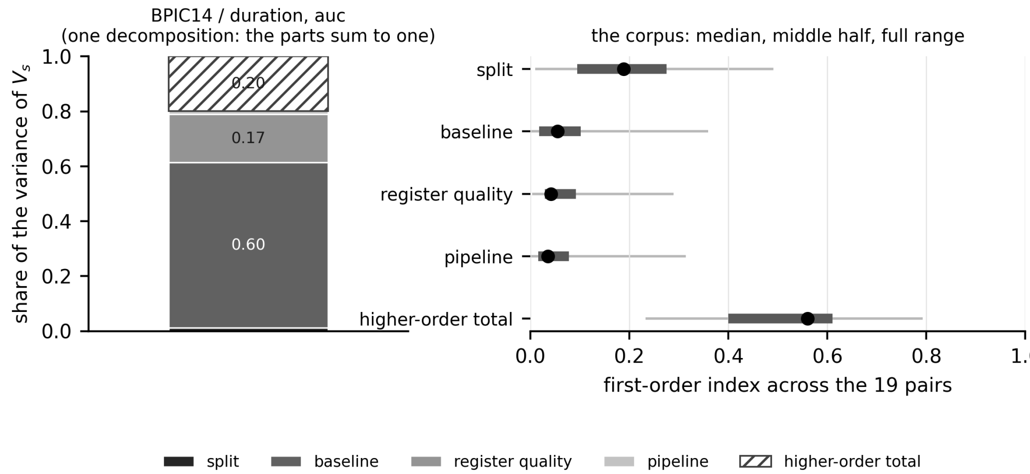

Figure 2: The decomposition, within instrument, under the equal-level measure. *Left*: one pair at one instrument, so the components are a decomposition and the parts sum to one exactly — the largest pair in the corpus, at the reference instrument, which is the cell every other table reports at. The hatched segment is the total higher-order share $1-\sum_i S_i$ of (3). *Right*: each axis's first-order index across the corpus, as the median over pairs with the range the middle half occupies and the full range. Nothing on the right is stacked, because a median over pairs and a median over instruments do not add and a stacked bar of them would invite the addition. The bootstrapped indices and their intervals are in Table S5.

$\max_i(S_{T_i} - S_i)$ is a different quantity and is 43.7% at the median pair.

The measure is not neutral, and one conclusion does not survive it. Across the three product measures the higher-order share moves by 24.3%, and under the measure concentrated on the reference cell it is 31.7% — smaller than the largest first-order index under the same measure, 39.2%, on 12 of 19 pairs. *The interactions carry more than any main effect* is therefore a claim about a measure and not only about a surface. Supplement S10.10 carries the target axis, which axis leads on which pair, and both invariance tests.

The baseline axis moves the answer by more than the answer. Holding the pipeline, the split, the register and the metric at the reference and varying only the baseline rung, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself at the reference cell on 15 of the 19 pairs. The index and the spread answer different questions, what share of the *variance* an axis explains while every other axis is also varying against how far that one axis moves the answer on its own, and both are reported because either alone misleads.

The pipeline axis is two axes. Its levels are a model family together with an encoding and, in one case, a calibration, and on 15 of the 19 pairs only two of them run. Crossed on the case study's log — both families against all three encodings on both targets, 12 complete factorial pipelines — **the encoding is the larger of the two**, 10.1% of the variance against the family's 5.2%, with an interaction, 9.7%, as large as either main effect: the fused index is not the sum of two indices. Supplement S10 carries the crossed decomposition. It has not been re-run on the other 15 pairs, which Section 11 records.

6.3. Resolution regions, and the coverage they have

Table S6 and Figure S4 carry the region labels and the robustness index for every pair, computed on the inference surface under the whole-surface simultaneous band at the conservative end of the critical value's Monte Carlo interval. That family has a median 180 members per pair and a median critical value of 3.33, against 2.35 for a within-instrument family and 1.96 pointwise; under the narrower family the same draws give a different region label on 9 of the 19 pairs, and the whole-surface family resolves 1,150 cells against the narrow family's 1,510.

The nominal critical value is not enough, and the correction is measured rather than assumed. Section 10.3 shows that coverage is governed by K/n and that this corpus lives where coverage is below nominal, the median pair sitting at 0.106, so a region label computed from a nominal band overstates how much of a surface is resolved. The (n, K) plane is dense enough to calibrate against, and Supplement S9 derives the inflation factor $c(K/n)$ from it — a monotone regression through the measured factors, whose log-linear summary has slope 0.102 (standard error 0.005). The factors the corpus draws range from 1.18 to 1.77; at the median pair the critical value goes from 3.33 to 4.54; the corpus resolves 896 cells against 1,150, a loss of 22.1%, and 4 of the 19 region labels change. **Every region label and every ρ quoted in this paper is the calibrated one**; Table S6 prints the nominal columns beside them.

The factor's reach, and what widening the plane in n found. It is fitted at training sizes 500 to 180,000 against a corpus running 735 to 176,213, so all 19 of the 19 pairs lie inside the range that identifies it. Two things took the place of the extrapolation that used to sit here. K/n **is not sufficient**: $\log n$ enters the fit at $t = 8.3$, and the applied curve, a function of the ratio alone, sits below the measured factor's own lower confidence limit on 2 of the plane's cells. **And on 4 of 33 cells the factor does not exist**, because past a crossing in n the basic interval stops containing the point estimate a widening would be about (Section 4.1). Supplement S9 carries the ladder, the two-covariate comparison and the admission rule; Section 11 carries what it leaves the corpus's largest pairs.

Of the 13 pairs that resolve anything, 9 have no harmful cell and 3 resolve cells of both signs. That is the informative reading of the region column, and it is the one to carry away: on two thirds of the pairs where the data determine any sign at all, every sign they determine points the same way; on the remaining third they do not.

No surface in this corpus is uniformly beneficial, and that statement is weaker than it looks — the label requires every admissible cell to resolve beneficial, and a cell with a negative point estimate can never do so at any critical value (Supplement S10.17). The counts are reported at four calibration settings and the ordering is stable across all of them (Supplement S9.7), so no conclusion here rests on the calibration being exactly right.

6.4. What one number costs

A conventional report — one pipeline, the clean register, the single holdout, ROC AUC — disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. Two things are wrong with reading that as *what a one-number report exposes a reader to*, and both are corrected before the headline is stated.

The axes are not all the same kind of thing. The design space mixes choices an analyst makes with things that are not choices at all, and Table S18 partitions the decomposition accordingly: *analyst latitude* is the pipeline and the baseline rung; *resampling* is the split, five of whose six levels are expanding-origin folds on the same data; *counterfactual* is the register-quality condition, a different state of the world rather than a different way of analysing this one. On the primary scale, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, and resampling is the larger of the first two on 10 of the 19 pairs.

Whether the split moves the increment more than the analytic axes depends on which scale the partition is taken under, and we report both rather than the one that reads better. The primary scale decomposes *inside* each instrument and takes the median across the five, so the instrument is a stratum and its own contribution is in none of the three columns. Making it a sixth axis needs the headroom rescaling of Section 4.5, because raw AUC and raw average precision do not share units — and on that scale the ordering reverses: analyst latitude carries 19.2%, resampling 9.4% and the counterfactual 2.7%, with resampling the larger on 7 pairs. The instrument is an analyst's choice, so the second number is the one that answers the question as asked; the first is the one taken on the units the rest of Section 6 uses. Neither is a corpus fact independent of the scale, which is the paper's own thesis applied to the paper. Restricted to the 30 analyst-latitude cells a pair carries, the disagreement rate is 20.0% against the pooled 32.8%: the first is what a one-number report exposes a reader to, the second what a claim ranging also over degraded registers and resampled splits is exposed to.

The reference specification is itself a choice, and moving it moves the headline by less than the corrections above. The rate counts cells whose sign disagrees with the reference cell, so the reference supplies the sign and nothing else. Varying it one axis at a time over every level that axis carries — 26 variants in all, Table S19 — the corpus median runs 32.8% to 38.4% against the declared 32.8%: 35.3% if the reference pipeline becomes target-encoded boosting, and 32.8% to 38.2% across the five instruments. The headline is not an artefact of which cell we called conventional.

And most cells are not cells whose sign the data determine. The median pair resolves 8 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction, so a rate computed over unresolved cells is partly measuring that.

On the cells the corpus can resolve, the rate is 7.5%. Of the 896 cells whose coverage-calibrated whole-surface band excludes zero, 67 disagree in sign with the reference cell. Every cell it counts is one where the data do determine a sign, and that sign contradicts the number a conventional paper would print.

Applying both corrections at once gives 12.9%, which is the number this section's own argument asks for. The two corrections above — restrict to the axes an analyst chooses, and restrict to the cells the band resolves — were made separately. Imposed together they leave 210 cells, of which 27 disagree. That is larger than the resolved-cell rate and smaller than the analyst-latitude rate, and it is the rate that answers "how often would a one-number report be wrong about a sign that the data determine, on a cell an analyst might actually have stood on".

6.5. Against specification-curve analysis as practised

The closest existing practice is specification-curve analysis [29]: enumerate the reasonable specifications, plot the curve, and test it against a sharp null by permutation. Table S20 runs it unchanged on *every* admitted pair — 19 of them, 100 permutations of a declared 24-cell sub-surface — beside the region label computed on the same cells, which makes "how often do the two verdicts part company" a measurement rather than an illustration. Two budgets are declared, the sub-surface and a cap of 20,000 cases per pair that binds on 7 of them; Supplement S10 gives both.

They part company on 9 of the 19 pairs, close to half. The permutation test rejects the sharp null — the surface is decisively unlike one with no signal — and the same cells contain increments of *both* signs. A reader given only the test would conclude that the register matters and have no way to learn that whether it helps or hurts depends on choices nobody told them about. On 3 pairs neither resolves and the cheaper instrument suffices; the rest are mixed.

And a specification curve is a statement about the cells it enumerates. On 1 pairs every cell of the sub-surface has one sign while the *full* surface is sign-changing or conditionally harmful. Nothing is wrong with either — they are claims about different sets — which is the point: without a declared denominator and a simultaneous band, a curve cannot distinguish "the sign holds" from "the sign holds on the cells I drew".

Neither subsumes the other. The permutation test answers *is there anything here*, costs one refit per permutation per cell and needs no band; the region answers *may I quote a sign*, costs a nested bootstrap, and is what a reader deciding whether to act needs. This paper's contribution is the second, and the first should be reported beside it.

6.6. Decision rules, in sample and out

The four rules of Section 4.7 are compared under the loss defined there, inside one instrument at a time. Out of sample — choosing on a random half of the admissible cells and paying on the other half — the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean. **Three results run against the surface's advocate and we report them**: the conservative rule is worst in 4 instruments of 5, the one-number rule's median excess regret is *zero* in every instrument, and under rolling-origin folds the ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number; the case rests on the decomposition and the sign disagreement, and this is a check on those. Table S25 and Supplement S10.8 carry it.

7. Case study: a configuration management database

A configuration management database is the register an IT estate keeps of the items it contains, and it is bought partly on the argument that operational analytics need it. This section evaluates one against a named decision on a public log: will this case be reassigned to another group before it is resolved? The estate is a bank's and the log is BPI Challenge 2014 [10]; the register field is the affected configuration item, with 3,019 distinct values. It is reported at greater length than the other 19 pairs because it is the case that motivated the study and the one on which the decision time — when in the service-desk process the prediction is made — decides the answer.

7.1. One log, two published answers, and what separates them

This log appears twice in this paper. Section 6 applies the registered generic rules of Section 5, which admit 46,606 cases and define the target by the registered handover rule. This section uses the cohort and the reassignment target defined for the estate: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The first set is a strict subset of the second; the extra 1,151 cases are those the estate's own cutoff excludes.

The two produce different answers about the register at the later decision time, and the difference is large enough to change the sign. Against intake plus group plus the knowledge reference, the estate's cohort and reassignment target give -0.0029 and the registered cohort and handover target give $+0.00768$, on which the planned contrast PC3's one-sided p-value rejects at the Holm-adjusted level while its interval does not exclude zero (Section 4.4). Table S8 takes the manuscript from the first to the second one factor at a time.

It is the target, not the cohort. Over the cohort $\times$ target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400; handover between activity groups, under the registered rule, has prevalence 0.927. They are not the same question and the register is not worth the same to them. Two further candidate explanations were checked and are not explanations: the fields reached by the two analyses agree on every row.

An axis nobody declares, and this paper was standing on it. Both analyses sort the log in time and split at seventy per cent of the row order, and 270 incidents share their timestamp with another. How those ties are broken decides which cases fall either side of the split, and the sort the case study originally used was not stable, so its tie order was neither declared nor reproducible: it printed −0.0032 where the two declared rules both give −0.0029. Over 25 random tie-breaks the increment at the knowledge rung ranges over 0.00088 AUC with a standard deviation of 0.00017, and it does not change sign (0 of 25 draws are positive), so no conclusion here turns on it — but it is a quarter of the point estimate's own magnitude, produced by a choice no reporting standard asks anyone to state. Every number in this section is computed under a declared tie-break, by incident identifier, on the estate's cohort and reassignment target, and every caption says so; the registered cohort's answers are in Section 6 with the other 19 pairs. The decision-time result below is therefore a statement about a routing decision on this estate, not about handover in general.

7.2. Three decision times

A decision time fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — and the estate's surface is estimated at each of three moments. **The register's worth turns on which one the analyst occupies**: at τ_0 it is +0.2310 [+0.2204, +0.2618] over 145,478 calls, at τ_1 +0.2013 [+0.1930, +0.2224], and at τ_2 against the intake block plus the group it is +0.1034 [+0.0763, +0.1115] while against the block, the group and the knowledge reference it is −0.0029 [−0.0078, +0.0015] — not resolvably different from zero. **The step from τ_1 to τ_2 is an information step only on the matched population**: 942 of the 45,455 incidents have no interaction record, so the fourth row of Table S9 restricts τ_2 to τ_1's own cases and it is that row, not the third, that isolates the information. Supplement S10.14 gives each moment's population, register and admissible set.

7.3. The leakage tipping point, and register quality

Two further experiments on this estate are in Supplement S10.4 and summarised here.

A field that looks like leakage has a tipping point, and where it sits is what a reader has to judge. The knowledge reference is the field that takes the register's increment to indistinguishable from zero, and whether it was available at τ_2 is probable but unproven. Writing λ for the share of its values assumed written *after* the incident — so that a larger λ makes more of the field inadmissible — the register is worth -0.0029 $[-0.0078, +0.0015]$ at $\lambda = 0$, not resolvably different from zero, and reaches half of its knowledge-free value at $\lambda = 0.64$. A reader who believes more than 64% of references are written after the incident should read this estate's register as worth at least half of +0.1034 AUC, and one who believes almost none are should read it as worth almost nothing. We cannot measure λ, so the disposition of an unprovable admissibility assumption is reported as a sensitivity analysis and not as a headline (Figure S2, Table S22).

Register quality is an axis, not a caveat, and which rows go missing matters as much as how many. Under six declared degradation mechanisms the increment at the reference cell moves from +0.1034 on a clean register to +0.0819 when half the population is lost from its long tail and +0.0708 when half is lost from its core. Three of the mechanisms are *dependent* — on time, on content, and on a training-estimated propensity that correlates missingness with the outcome — and they carry severity integrals +0.0768, +0.0486 and +0.0559 against +0.0692 for the independent long-tail mechanism, which they bracket rather than track. A quality axis built only from independent coin flips measures one corner of the question. Tables S29 and S30 carry every mechanism and level.

7.4. What this says to a configuration management programme

The estate's own reading is in Supplement S10.12: what the register is worth here, at which decision time, under which target, and what a programme would have to change for that answer to move. It is a reading of one estate and is offered as one.

7.5. The absorption, and why the ratio is secondary

The quantity a report of this kind usually leads with is the *relative* reduction R: the share of the register's apparent value that a single free field absorbs. The absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ is reported first, with its own interval, and R second with a Fieller set and the set's kind. On this estate at the reference cell D is +0.0804 [+0.0626, +0.0914] and R is 0.437 with a bounded Fieller set; 0 of the 30 reductions this ladder produces have a Fieller set that is not a bounded interval, and for those a percentile interval is not a summary of anything. Across the corpus the figure is 5,282 of 10,584, and the full ladder is Table S31.

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit, which is: the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. The operating point is a cost ratio, not a knob

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\mathrm{NB}(\theta) \;=\; \frac{\mathrm{TP}(\theta)}{n} \;-\; \frac{\mathrm{FP}(\theta)}{n} \cdot \frac{\theta}{1-\theta},$$

so $(1-\theta)/\theta$ is the number of false nominations the desk is willing to accept for one true one. Choosing θ is choosing an operational cost ratio, and it belongs to the analyst's declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$, or 9.0 false nominations accepted per true one, where a review is cheap and the desk nominates liberally; 0.30, or 2.3, where a review and a miss cost about the same; and 0.60, or 0.7, where review capacity is scarce. A reader whose desk sits at a ratio not in that list reads their own point off Figure S3: the operating point is the reader's to choose, so the analyst reports the curve rather than a point on it.

8.2. Calibration comes first, or the threshold means nothing

A threshold is a statement about probability, so a model whose probabilities are wrong is being read at a threshold it does not have. Every decision curve here is therefore computed on *calibrated* scores, with the calibrator fitted inside the training half and rebuilt inside every bootstrap refit, and a registered rule excludes an arm whose calibration slope leaves $[0.5, 2]$. Supplement S10.5 gives the rule, what it removes and what the curves look like without it.

8.3. Simultaneous bands change what can be said

The decision-curve family is declared separately from the scalar surface and carries its own max-t critical value, because net benefit and a scalar instrument are not the same scale. Across the 31 operating points of this estate's curve **the pointwise interval declares the increment resolvably positive at 24 of them and the simultaneous band at 18**, and the harmful counts are 0 against 0. Both are built from the same draws, so the only thing that differs is the family: those points are the price of a statement about a *range* of thresholds rather than about each one separately, and a paper that reads a decision curve across its range while pricing it pointwise has claimed the first and paid for the second.

This family is also the one most exposed to how the critical value was estimated. It is *not* coverage-calibrated — the (n, K) calibration of Section 6.3 is estimated for the scalar family — and Section 10.4 finds that the widening it would need is about twice what the surface families need. Supplement S10.6 carries the comparison, the dip near $\theta = 0.55$ and Figure S3; Table S23 prints the curve.

8.4. The decision, in units a service desk uses

The comparison is run in net benefit per thousand cases, over a declared distribution of exchange rates a desk plausibly holds (0.08 to 0.36, centred 0.19), and only on models that pass the registered calibration rule — which excludes 40 of 118 arms and 3 of 19 pairs. **That count is itself the strongest operational finding here**: a decision curve read off a model whose probabilities are wrong is read at a threshold the model does not have.

And the rules mostly tie. The one-number rule's expected excess is 0.000 per thousand against the conservative rule's 0.048, the four rules separate on only 11 of 19 pairs, and the one-number rule is worst on 10. **The case against one-number reporting does not rest on this design.** What does separate in this unit is the decision time: the register promises 1.71 per thousand at the earlier moment and delivers 0.003 at the later one, a shortfall of 1.70. Supplement S10.9 carries the rules, the exchange-rate distribution and the exclusions; Tables S21 and S25 the numbers.

9. A reporting standard, and software

9.1. The minimum reportable form

The apparatus of Sections 3 and 4 implies a reporting standard short enough to state in full. A feature-value claim is reportable when it carries: **the design space**, as an enumerable declaration of which learners, encodings, targets, splits, decision times, quality mechanisms, baselines, metrics and operating points are admissible, why anything defensible was left out, and *the measure placed on those levels*; **the denominator**, as a count a reader can reproduce from the axis levels — declared cells, computed cells, and the cells any inference actually rests on, which need not be the same set; **the absolute increments** at every rung of the baseline ladder, in the metric's own units, with intervals from a resampling scheme that refits the model; **the sensitivity decomposition**, with the total higher-order share reported as $1 - \sum_i S_i$ and not as any per-axis difference; **the resolution region and** ρ, under a band whose family is the set the label ranges over; and **the ratio** R **last if at all**, with a Fieller set and the set's kind, never where the denominator is not resolvably positive. Nothing in that list is expensive once the surface exists, and the surface is a factorial over declarations the analyst has already made implicitly.

9.2. `fieldvalue`

The standard ships as `fieldvalue`, which computes the surface from a data frame, a feature name and a dictionary of baselines. Its notable property is a refusal: `Surface` raises rather than returning a scalar, `Surface.report()` emits the form above, and `Surface.at(...)` returns one cell only if the caller names every axis. Version 0.2.0 adds the four objects of Sections 4.5–4.7 as functions of a surface — `decompose`, `regions`, `robustness`, `regret` — each taking a tidy frame rather than this package's own object, so they apply to a surface computed by some other pipeline. It carries 108 tests and is an *independent implementation*: the analysis scripts and the package compute the same four objects from the same definitions in different code, neither written against the other, and on this corpus they agree on 266 of 266 quantities to 1.1×10^{-16}.

9.3. Beyond process logs, and beyond this paper's claims

The standard is not about event logs and not about configuration databases. Supplement S7 runs it on UCI Adult, where the expensive field is `occupation` and the cheap ones come with the sampling frame: over 480 cells the reduction runs 0.200 to 0.947; the baseline and the metric carry the same share of the variance to three figures, 25.9% and 25.9%, against 7.2% for the register population; and a one-number report misstates the sign for a mean 24.9% of the other cells. `examples/worked_example.py` produces all of it in 121 statements against the package's public interface.

How the surface is reported today. Whether the axes this paper varies are varied in published work deserves a systematic review, and we did not conduct one. A machine-assisted prevalence pilot was run on a convenience frame and is *not* reported here: with one adjudicator, no independent human reliability study and a mechanical screen whose measured sensitivity is 0.474, it can generate a hypothesis about practice and cannot test one. Its frame, sample, codes, adjudications and evidence dossiers are in the archive. The paper's argument does not need it: the case against one-number reporting is made by the decomposition and the sign disagreement, on data, and not by a count of how often other authors report one number.

10. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the estimator where it is. The full tables are in Supplement S11 and the diagnosis of the one world that failed is in Supplement S6.

10.1. Six worlds, two estimands

The simulation uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each, with the covariate space finite so that the population AUC of any scoring rule is computed in closed form rather than simulated. Two estimands are separated because under misspecification they differ: V_{oracle}, the increment between the two Bayes-optimal rules, and V_{limit}, the increment between the two fitted pipelines' limits. A resampling interval can only cover the second, and reporting its coverage of the first as the estimator's coverage is a category error. Supplement S6.1 describes the worlds and the marginalisation a corrupted register forces; Table S35 reports both estimands.

10.2. Coverage, bias and width

Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points at 1,000 replicates, which is the resolution every comparison here should be read at; Figure S5 draws them by construction and world.

Where the refit helps, and where the percentile interval fails. The nested basic interval covers a median 93.0% against the fixed-model interval's 92.1%; on the sparse world the *percentile* interval collapses to 37.2% and the basic construction restores it to 89.9%, which is why every interval and band here is the basic one (Supplement S6.2; Table S35).

What no interval repairs. Against V_{oracle} rather than V_{limit}, every construction undercovers on the drifting world — 65.8% at best — and it should: when the coefficients move between eras the estimator is not consistent for the oracle quantity, and the largest gap is 0.0161 in AUC units. Under misspecification an interval around the estimator's own limit is not an interval around the quantity a reader cares about, and no resampling scheme repairs that.

10.3. Sample size, register cardinality and block length

A construction that works at one sample size, one register cardinality and one block length has been validated at a point, not in a regime. Sample size and register cardinality are therefore varied *jointly*, up to $K = 3{,}019$ levels — the cardinality of the case study's own register — and the block length is varied around its rule of thumb.

The ratio K/n captures most of the dependence, and not all of it. Across the plane the basic interval covers a median 83.3% and a minimum 21.3%, the last at a ratio of 0.377, and it is short of nominal everywhere. It is not sufficient: two cells at nearly the same ratio cover 0.780 and 0.613, a gap of 3.2 combined Monte Carlo standard errors, and holding the register fixed while moving the training half from 2,000 rows to 8,000 moves coverage from 0.453 to 0.613. **This corpus sits in the regime where that matters**: its median pair has $K/n = 0.106$ and 10 of 19 pairs are above a tenth, so a nominal band on them is *narrower* than a 95% interval should be. Section 6.3 calibrates against the plane and Supplement S9 gives it in full.

Where the interval stops containing its own estimate. The denser plane the coverage calibration of Section 6.3 is fitted on — a different and larger object from the 10-cell grid above, at 500 to 180,000 training rows — locates the crossing Section 4.1 describes. At a fixed ratio of 0.125 the basic interval straddles its own point estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000, because the bootstrap displacement is a bias that does not shrink with n while the half-width does. Past that crossing the inflation factor is not defined — it is a widening about the estimate, and there is nothing to widen about — and 4 of the plane's 33 cells have none for that reason, every one of them at 50,000 rows or more. Section 6.3 reports what that leaves the corpus, and Supplement S9 describes that plane's design.

Block length. 3 lengths are examined — half the rule $\lceil n^{1/3} \rceil$, the rule, and twice it. Coverage moves with the length, spanning 3.6% within a world, of the order of two standard errors; but the comparison the paper rests on is unchanged at every length, the basic interval beating the percentile interval on the sparse register by at least 46.4% at all 3.

10.4. The band's family-wise coverage

Everything above validates a *pointwise* interval. Two of this paper's four reporting objects are functions of a *simultaneous* band, and until this section nothing measured what that band covers.

The design, and why its answer is an upper bound. The truth is zero at every cell of a synthetic family. One draw from the sampling distribution gives the point estimate, and as many further draws *from that same distribution* as the corpus family it is matched to carries give the bootstrap — so the bootstrap is not an approximation of the sampling distribution but is it. Every way a resampling scheme can be wrong — dependence it fails to reproduce, a refit that comes apart, a block length misjudged — is switched off, and what is left is the object under test: a standard error and a critical value both estimated from the same handful of draws. A coverage measured here is therefore the most the real procedure can attain, and a shortfall here is one that no improvement to the resampling repairs. The families are matched to this corpus in size, draw count, cross-cell correlation, effective rank and per-cell excess kurtosis; the matching is fitted by grid search against the corpus's own draw files and reported beside its target rather than asserted. Coverage is over 2,000 replicates in each of 18 cells, so no coverage here is determined worse than 1.1%.

Three regimes, because "the draws are heavy-tailed" turned out to name two different things. Under *Gaussian* draws the multiplier's family-wise coverage is a median 87.3% and a minimum 74.8%. Under draws carrying this corpus's tails, whose median cell has an excess kurtosis of 0.32 on the surface families and whose ninetieth percentile is 3.6, the median is

84.6% and the minimum 76.1%. Under the third regime, which adds the degenerate cells described below, they are 84.4% and 39.2%.

What is wrong with the decision-curve families is not a tail. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic, and 841 cells of the corpus's curve families are exactly zero in at least nine draws of ten, against 0 anywhere in the surface families. They are what makes q_{emp} large there; excluding them moves the median ratio q_{emp}/q from 2.09 to 1.80 and does not dissolve the disagreement. Supplement S9.6 carries the mechanism and Table S38 the regime with and without them.

How far short, in the units the calibration works in. A critical value is widened multiplicatively, so the useful summary of the shortfall is the factor that would have closed it. On the surface families that factor is 1.05 to 1.24; on the decision-curve families, 1.90 to 2.05. The coverage calibration of Section 6.3 applies 1.09 to 1.88, is fitted to restore *pointwise* coverage rather than this one, and is applied to the surface families only — so it is of the right order where it acts and absent where the shortfall is largest. Section 11 says what follows.

The draw count is the binding constraint. At the modal surface family the multiplier band covers 82.0% at 33 draws and 96.6% at 1,000, and the per-cell interval underneath it moves 90.3% to 94.9% over the same range: part of what the band loses it loses before any critical value is chosen, and Section 10.2 validated that construction at a refit count this corpus does not reach on most pairs. No estimator recovers the difference. The best of the five, the empirical quantile at the upper end of its own order-statistic interval, reaches 92.2% at 1.52 times the width and still falls to 86.5% at the smallest family; a Rademacher wild multiplier, which preserves the observed draws' magnitudes instead of replacing them by Gaussians and is the natural repair for a tail problem, is worse than the Gaussian multiplier at 80.7%.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a plane of 10 (n, K) cells and 3 block lengths, and the band on the 18 matched families of Section 10.4 — on nothing else. The stationarity and mixing the moving-block bootstrap needs are assumed and not tested; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about.

11. Limitations

This is predictive performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a prediction changes an outcome, or what a data programme is worth. The quantity is an increment in predictive performance, read at a declared exchange rate in Section 8. A reader who converts Table 3 into a procurement decision is doing something this evidence does not support.

The design space is declared, not complete, and one omission is structural. $\mathcal{S}$ has nine axes here. It could have more — hyperparameter tuning protocol, missing-data treatment for the baseline fields, the horizon of the target, the label-noise model — and the cohort rule is varied only on the case study's log, so the honest form of the claim is "the increment moves this much over *this* space". The structural omission is that **prefix length, prefix bucketing and sequence encoding are not axes here and cannot be**, because this study predicts at one prediction point from creation-time attributes. Those are the axes predictive process monitoring varies most, so the corpus result measures sensitivity in a simpler adjacent setting on the same data rather than in that literature's own benchmark, and adding a prefix axis on at least one log is the clearest next study this paper implies.

The axes are not all the same kind of thing. Three of the nine are choices an analyst makes, one is a resampling scheme and one is a counterfactual state of the register, and a variance decomposition that pools them answers a question nobody asked. Section 6.4 partitions them and reports the sign-disagreement rate on the analyst-latitude sub-space (20.0%) beside the pooled one (32.8%). The region labels and ρ are *not* partitioned, and are therefore conditional on the quality axis being averaged over: a surface is asked to be resolvably beneficial in cells where three quarters of the register has been deleted, which is part of why no surface here is uniformly beneficial.

The inference surface is smaller than the declared surface. Every point estimate comes from the full declared space; every *band* comes from the inference surface, a median 12.5% of each pair's computational surface, and the draw counts run 40 to 150 as declared. Supplement S8.1 gives what that budget costs and where it binds.

Every weighted quantity depends on a measure that is a modelling choice. A decomposition, a robustness index and a regret are all taken with respect to a distribution over specifications, and there is no neutral one. Three product measures and an envelope over a fourth are reported (Table S24); across the three the total higher-order share moves by 24.3%. **One qualitative conclusion does not survive all four**: under a measure concentrated on the reference specification, a single axis carries more variance than everything higher-order combined on 12 of 19 pairs, so "the interactions dominate" is a claim about a measure and not only about a surface. Two structural invariances, to duplicating a level and to refining a grid, are enforced and tested, so no number here can be moved by re-describing the same design space. Separately, the decomposition inherits the metric's scale (Proposition S1), which is why the within-instrument decomposition is primary and the cross-instrument one is a sensitivity analysis.

One estate for the case study, and a heterogeneous corpus. The configuration management database analysis is one bank's estate, on one public log, and the availability question at its centre is settled by probability rather than by a timestamped field-value history; the strongest version of this study would have an organisational partner supplying timestamped field histories, a known prediction time, historical register population and accuracy, and operational review costs. The 19 pairs are also not replications of one another: across the 13 logs the two registered targets label the same case identically on a median 51% of cases. Every corpus-level statistic is therefore about *sensitivity*, and every one is reported on the ITSM family as well as on the whole corpus (Table S15).

The quality mechanisms are synthetic. Population loss, identity error, reconciliation failure and staleness are constructed from the training half, not observed. Three of them are *dependent* — on time, on the item's recorded type, and on a training-estimated propensity that correlates missingness with the outcome — which is closer to how a register degrades than an independent coin flip, and every one is verified by execution to read no test outcome. But a constructed mechanism is not an observed one, and validating them against a genuinely incomplete register is the obvious next study.

The coverage calibration is no longer an extrapolation in n, and what replaced that problem is worse. All 19 of the 19 pairs now lie inside the range the factor is identified on, 500 to 180,000 training rows. Two things took the place of the extrapolation. **The applied curve is misspecified, measurably**: $\log n$ enters the fit at $t = 8.3$, the curve is a function of the ratio alone, and it sits below the measured factor's own lower confidence limit on 2 of the plane's cells, both at large n; a two-covariate fit does not repair it (Supplement S9). **And on 4 of the plane's 33 cells the factor does not exist**, because past a crossing in n the basic interval no longer contains the estimate a widening would be about. **This is not confined to the simulation**: on the corpus 74 of the 3,900 whole-surface cells have a band that excludes their own point estimate, on 3 pairs and reaching 21.1% of the cells on the worst. For those the calibrated band widens an interval the calibration's own estimator could not have been fitted on, and we do not know what it covers.

The interval construction was chosen on evidence from one sample size, below the size at which it misbehaves. Section 10.2 compares seven constructions and selects the basic one, and that comparison runs at 4,000 training rows and at no other size. Section 4.1 reports that the same construction produces a band excluding its own point estimate once the bootstrap displacement exceeds the band's half-width, which the plane places above 8,000 rows at a ratio of 0.125 — so the evidence that chose the construction was gathered below the regime in which the construction develops the defect. Whether the percentile, the bias-corrected or the m-out-of-n interval behaves better up there is not known: none of them was priced above 4,000 rows either. Re-running that comparison across the plane's range in n is the first thing we would do, and it is cheap relative to the surface.

The band does not attain its nominal level, and this is the sharpest limitation in the paper. Section 10.4 measures family-wise coverage against a known answer over families matched to this corpus. The multiplier band covers a median 84.6% and a minimum 76.1% against a nominal 95% — 87.3% and 74.8% even under Gaussian draws, the case the construction is derived for — and the bootstrap in that experiment is *ideal*, so those are upper bounds on what the procedure attains here. The safeguard we described as conservative, resolving only at the upper end of the critical value's Monte Carlo interval, moves coverage by less than a point: it controls the Monte Carlo error in q, not the error in what q estimates. **The repair is more draws, not a better critical value** — coverage runs 82.0% at 33 draws and 95.8% at 400 — and a draw refits the whole design space, which is why this version does not have them. We report the bands as computed rather than substituting a wider estimator that is also wrong and merely less visibly so.

What that leaves the two families is not the same. The multiplicative widening that would have given nominal coverage is 1.05 to 1.24 on the surface families, and the calibration of Section 6.3 already applies 1.09 to 1.88; it was fitted to restore *pointwise* coverage and not this, so that it is of the right order is fortunate rather than by design, and we claim no more than the order. On the decision-curve families the widening needed is 1.90 to 2.05, about a doubling, **and the calibration does not touch them at all**. Those conclusions are the ones to distrust. Neither statement is a coverage guarantee and this version does not have one.

Some of what the decision-curve comparison counts is arithmetic. 841 cells of those families are zero in at least nine draws of ten, because at those thresholds both arms treat every case alike, and they inflate q_{emp}: excluding them moves the median ratio q_{emp}/q from 2.09 to 1.80. A reader who prefers the empirical critical value should therefore read Section 4.3's second figure of each pair on the surface families and not on these.

Inference still rests on further assumptions. **The split point is fixed within a draw**, so split-position uncertainty is outside every interval in this paper. **The moving-block construction needs stationarity and mixing** that an event log is not guaranteed to have; neither is tested here. **The max-t band's family-wise coverage is asymptotic**, as is the multiplier bootstrap that estimates its critical value, and Section 10.4 shows what that asymptotics is worth at the 40 to 150 draws this corpus runs at. The block length is a convention, $\lceil n^{1/3} \rceil$; Section 10.3 varies it and the coverage of the reported interval does move with it, by up to 3.6% within a world, while the comparison the paper rests on does not.

The band's coverage is calibrated, and the calibration is itself an assumption. The inflation factor of Section 6.3 corrects the *scale* of the studentised statistic, which is the channel the (n, K) plane identifies. It does not correct a change in the shape of the maximum's distribution; it is estimated on a pointwise interval and applied to a simultaneous band; and it assumes the simulated worlds' dependence on K/n transfers. A reader who rejects the transfer should read the region labels and ρ as illustrative and rely on the objects that need no band: the point estimates, the decomposition, the all-cells sign-disagreement rate and the regret comparison.

The pipeline axis is crossed on one log only. Section 6.2 separates the model family from the encoding on the case study's log and finds the encoding axis the larger of the two. The other 15 pairs carry the two confounded, so their pipeline index remains a mixture, and whether the split found on one log transfers is not established here.

We do not measure reporting practice. The prevalence of one-number reporting in the published literature would be worth knowing and this paper does not establish it. The pilot that attempted it is in the archive with its own sampling error, coder error and resolution; it is a hypothesis, not evidence, and nothing here depends on it.

12. Conclusion

The incremental predictive performance of a recorded field is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, a decomposition of the surface's variance into the contribution of each choice, simultaneous confidence bands from a bootstrap that refits the model and whose critical value is calibrated for the register's cardinality, region labels that say when a sign is robust, and a decision-theoretic benchmark that measures what one number gets wrong.

On every pair, at least one admissible specification makes the register worth less than nothing. That is a statement about point estimates and needs no band; what the bands add is where a sign is *determined*, and of the 13 pairs that resolve any sign at all, 3 resolve both. Four further results are worth carrying away.

The choices move the answer, and no one of them is the culprit. Varying the baseline alone moves the increment by 0.073 AUC at the median log–target pair, but the largest first-order index at the median pair is 28.7%, the total higher-order share is 56.0%, and which axis leads is a property of the pair. An analyst cannot learn from somebody else's paper which of their own choices their answer turns on.

The observed sign depends jointly on the data, the target and the specification. On 18 of 19 pairs, between a tenth and nine tenths of the admissible cells are positive, and a conventional one-number report misstates the sign for a median 32.8% of the rest, and for 7.5% of the cells the corpus can resolve — a pooled ratio, driven by the pairs Section 5.4 names as construct-validity threats, and 3.5% on the 8 pairs whose register is a maintained one. Only one of the three arguments is conventionally reported.

There is a defensible collapse, and choosing between collapses is not where the value is. Lemma S1 identifies the decision-optimal one — the weighted mean increment under a declared measure — but out of sample its advantage over the conventional report is small (0.00002 against 0.00450 AUC of excess regret), it reverses under rolling folds, and on the calibrated operational utility of Section 8.4 the rules mostly tie. What the decision-analytic reading does establish is sharper: on 40 of 118 models a threshold cannot be read as a cost ratio at all, and a desk that adopted this register on its *first-touch* number and deployed it at incident creation would be promised 1.71 true positives per thousand cases and receive 0.003.

The decision time is an argument of the estimand, not a preliminary. It fixes which cases are in scope, what is known about them, and which fields are admissible. On the case study, moving the prediction from first touch to incident creation costs -0.0297 AUC through the population, a further -0.0099 through the information once the population is held fixed, and -0.0076 more through the incidents that have no interaction record at all — before any baseline changes; and the same log's answer changes sign when the target changes, with 99.3% of the difference on the target and 0.0% on the cohort.

The practical recommendation is narrow and cheap once the surface exists: report the surface, the decomposition, the region and ρ; state the decision time and the target as parts of the estimand; make the absolute increments primary; use a ratio last, with a Fieller set, and only where its denominator is resolvably positive.

CRediT authorship contribution statement

Sudhir Dixit: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author has no affiliation with, and received no support from, any vendor of configuration management, IT service management or process mining software.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used a large-language-model assistant, Anthropic Claude, for three purposes. The model identifier and the date of every session in which it was used are recorded in `AI-USE.md` in the archive, because a disclosure that names a product and not a version is not reproducible. The identifier for the final analysis and revision round is `claude-opus-5` (2026-08-24).

1. **Software.** The analysis scripts, the `fieldvalue` package and the verification harness were written with the assistant's help. Every result in this paper is produced by that code, the code is public, and `REPRODUCE.md` regenerates every table and figure from the raw data by one command.
2. **A use in a study that is in the archive and not in this paper.** A literature-prevalence pilot, described in Section 9.3 and reported nowhere in this paper or its supplement, was adjudicated by the author with the assistant's help from evidence dossiers extracted from full text. That is one machine-assisted adjudicator rather than two independent human raters, which is among the reasons the pilot carries no claim here. Its dossiers are in the archive; the model identifier used for that adjudication was not recorded at the time and is stated as unrecorded in `AI-USE.md`.
3. **Writing.** The assistant was used to draft and edit prose, which is a use beyond the language-and-readability editing an author may make without further comment, and is disclosed here for that reason. The author directed the structure and the argument, verified every claim against the result files, and takes full responsibility for the content of the publication.

No generative model produced, imputed, augmented or selected any datum, any result or any citation. The reference list was checked against the published records.

Data availability

All data are public. Event logs are obtained by DOI: BPI Challenge 2014 (https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35), BPI Challenge 2013 incidents (https://doi.org/10.4121/uuid:500573e6-accc-4b0c-9576-aa5468b10cee), and the further public logs whose DOIs and SHA-256 checksums are listed in `data/corpus/CHECKSUMS.txt` — 26 files are downloaded and checksummed in all, of which the registered role-assignment rules of Section 5 admit 13, yielding the 19 log–target pairs analysed here. The counts are different quantities and Table S10 accounts for every downloaded log that is not among the admitted pairs, in one of two ways: with the registered exclusion code the rules gave it, or, for the 1 logs downloaded as a held-out set for the out-of-corpus test, with the code HELD_OUT and the statement that they were never offered to the admission rules. The literature pilot's frame, sample, codes, adjudications and evidence dossiers are in the archive; the retrieved full texts are other publishers' copyrighted PDFs and are not redistributed, but `scripts/s06_audit2.py --fetch` re-retrieves them from the DOIs in `results/s06_sample.csv`. All derived results (`results/*.csv`), all figures and all code are in the archive below.

Code availability

Code, derived results, the pre-registered protocols, the amendment log and the verification harness are archived at the archived release cited in the data-availability statement (DOI reserved, inserted at proof) (release v25.0) and developed at https://github.com/sudhirdixit1/cmdb-routing-queue-baseline. The archive contains a `Dockerfile` and a `requirements.lock` pinning every transitive dependency; `REPRODUCE.md` gives the exact commands and the expected runtime.

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